

KENDRIYA VIDYALAYA SANGATHAN
AGRA REGION
FIRST PRE-BOARD EXAM 2024-25
CLASS: XII
SUBJECT- PHYSICS(THEORY)

Max. Marks : 70

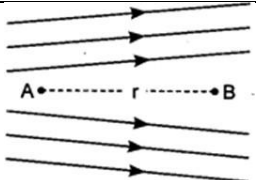
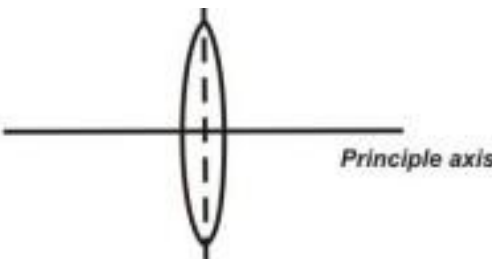
Time: 3 hours

General Instructions:

- (1) There are 33 questions in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- (3) All the sections are compulsory.
- (4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study based questions of four marks each and Section E contains three long answer questions of five marks each.
- (5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- (6) Use of calculators is not allowed.
- (7) You may use the following values of physical constants where ever necessary
 - (i). $c = 3 \times 10^8 \text{ m/s}$
 - (ii). $m_e = 9.1 \times 10^{-31} \text{ kg}$
 - (iii). $e = 1.6 \times 10^{-19} \text{ C}$
 - (iv). $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$
 - (v). $h = 6.63 \times 10^{-34} \text{ Js}$
 - (vi). $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$
 - (vii). Avogadro's number = 6.023×10^{23} per gram mole

SECTION : A

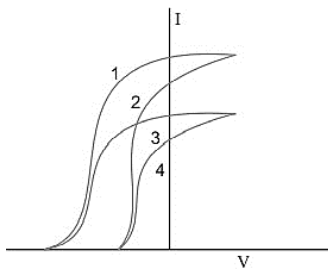
Q.NO.		MAR KS
1.	Dielectric constant for a metal is (a) zero (b) infinite (c) 1 (d) 10	1
2.	Kirchhoff's law of junctions is also called the law of conservation of (a) Energy (b) Charge (c) Momentum (d) Angular momentum	1
3.	Figure shows the electric lines of force emerging from a charged body. If the electric field at A and B are E_A and E_B respectively and if the displacement between A and B is r then	1

	 (a) $E_A > E_B$ (b) $E_A < E_B$ (c) $E_A = E_B / r$ (d) $E_A = E_B / r^2$	
4.	An electron is released from rest in a region of uniform electric and magnetic field acting parallel to each other, the electron will (a) move in a straight-line (b) move in a circle (c) remains stationary (d) move in a helical path	1
5.	The ground state energy of hydrogen atom is -13.6eV. What is the potential energy of electron in this state? (a) 0 eV (b) -27.2 eV (c) 1 eV (d) 2 eV	1
6.	A biconvex lens of glass having refractive index 1.47 is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is (a) 1.47 (b) 1.62 (c) 1.33 (d) 1.51	1
7.	A double slit interference experiment is carried out in air and the entire arrangement is dipped in water. The fringe width (a) Increase (b) Decrease (c) remains unchanged (d) fringe pattern disappear	1
8	What will be the de-Broglie wavelength of an electron accelerated from rest through a potential difference of 100 volts? (a) 12.27 Å (b) 1.227 Å (c) 0.1227 Å (d) None of these	1
9	The electromagnetic wave used in RADAR navigation is- (a) Gamma Ray (b) Micro wave (c) Radio wave (d) X - Ray	1
10	An equiconvex lens of focal length 15 cm is cut into two halves as shown in figure. Find the focal length of each part?  (a) -30cm (b) -20cm (c) 30cm (d) -15cm	1
11	The direction of the magnetic field at the center of a current-carrying circular loop is: (a) Along the axis of the loop	1

	(b) Parallel to the plane of the loop (c) Radially inward (d) Tangential to the loop	
12	In an LCR circuit the voltage across L, C and R is 10V each. If the inductor is short circuited, the effective voltage becomes (a) 10 V (b) $10\sqrt{2}$ V (c) $20\sqrt{2}$ V (d) 20 V	1
13	Directions: Each of these questions contain two statements, Assertion and Reason. Each of these questions also has four alternative choices, only one of which is the correct answer. You have to select one of the codes (A), (B), (C) and (D) given below. (A) Assertion is correct, reason is correct; reason is a correct explanation for assertion. (B) Assertion is correct, reason is correct; reason is not a correct explanation for assertion (C) Assertion is correct, reason is incorrect (D) Assertion is incorrect, reason is incorrect Assertion: In a non-uniform electric field, a dipole will have translatory as well as rotatory motion Reason: In non-uniform electric field a dipole experiences a force as well as torque.	1
14	Assertion: At a fixed temperature, silicon will have a minimum conductivity when smaller acceptor doping. Reason: The conductivity of an intrinsic semiconductor is slightly higher than that of lightly doped p-type.	
15	Assertion : In astronomical telescope, the objective lens is of larger aperture. Reason: Larger is aperture, smaller is the magnifying power.	1
16	Assertion : Average power in an A.C circuit having L only is Zero. Reason : Power factor of A.C circuit having L only is zero.	1

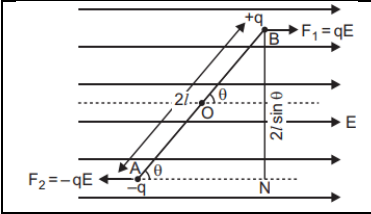
SECTION: B

17	Define magnetic flux. Write its SI unit. Scalar or vector	2
18	In half-wave rectification, what is the output frequency if the input frequency is 50 Hz? What is the output frequency of a full-wave rectifier for the same input frequency?	2
19	Show that the radius of the orbit in hydrogen atom varies as n^2 , where n is the principal quantum number of the atom. OR Two nuclei have mass numbers in the ratio 1: 2. What is the ratio of their nuclear densities?	2
20	The potential difference applied across a given conductor is doubled. How will this affect -	2

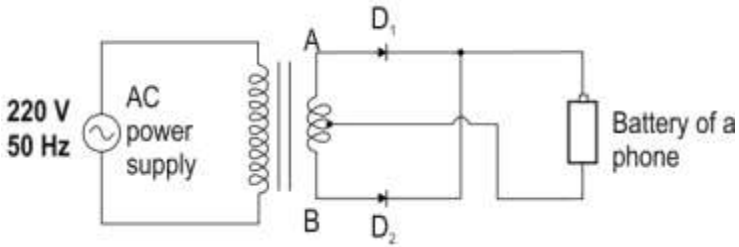
	(i) the mobility of electrons and (ii) the current density in the conductor? Justify your answers.	
21	A biconvex lens made of a transparent material of refractive index 1.25 is immersed in water of refractive index 1.33. Will the lens behave as a converging or a diverging lens? Give reasons. OR The power of a thin lens is +5 D. When it is immersed in a liquid, it behaves like a concave lens of focal length 100 cm. Calculate the refractive index of the liquid. Given refractive index of glass = 1.5.	2
SECTION C		
22	The given graph shows the variation of photo-electric current (I) with the applied voltage (V) for two different materials and for two different intensities of the incident radiations. Identify and explain using Einstein's photo electric equation for the pair of curves that correspond to (i) different materials but same intensity of incident radiation, (ii) different intensities but same materials. 	3
23	Arrange the following E.M. Wave, in increasing order of frequency and among these waves, which wave has greatest penetration power and ionization power? (a) X-ray (b) Gamma ray (c) Radio wave (d) visible light	3
24	The ratio of intensities of maxima and minima in an interference pattern is found to be 25:9. Calculate ratio of light intensities of the sources producing the pattern.	3
25	State Ampere's circuital law. Derive an expression for magnetic intensity due to a current carrying straight wire of infinite length.	3
26	Explain the term 'drift velocity' of electrons in a conductor. Hence obtain the expression for the current through a conductor in terms of 'drift velocity'.	3
27	Using Gauss's law, prove that the electric field at a point due to a uniformly charged infinite plane sheet is independent of the distance from it. How is the field directed if (i) the sheet is positively charged, (ii) negatively charged?	3
28	A series LCR circuit with $R = 20 \Omega$, $L = 2H$ and $C = 50 \mu F$ is connected to a 200 V A.C. source of variable frequency- (i) What is the amplitude of the current and its rms value. (ii) what is the average power transferred to the circuit in one complete cycle at resonance (iii) Calculate the potential drop across the capacitor? OR	3

	In an ideal transformer, the number of turns of primary and secondary is 1000 and 2000 respectively. (i) If maximum voltage in primary is 120V, what is the maximum voltage in secondary? (ii) if the current in primary coil is 5 Amp calculate current in secondary coil?	
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SECTION: D

29	When electric dipole is placed in uniform electric field, its two charges experience equal and opposite forces, which cancel each other and hence net force on electric dipole in uniform electric field is zero. However, these forces are not collinear, so they give rise to some torque on the dipole. Since net force on electric dipole in uniform electric field is zero, so no work is done in moving the electric dipole in uniform electric field. However, some work is done in rotating the dipole against the torque acting on it.  (i) The dipole moment of a dipole in a uniform external field $\vec{E}$ is $\vec{p}$. Then the torque τ acting on the dipole is (a) $\tau = \vec{p} \times \vec{E}$ (b) $\tau = \vec{p} \cdot \vec{E}$ (c) $\tau = 2(\vec{p} + \vec{E})$ (d) $\tau = (\vec{p} + \vec{E})$ (ii) An electric dipole consists of two opposite charges, each of magnitude $1.0 \mu\text{C}$ separated by a distance of 2.0 cm. The dipole is placed in an external field of 10^5 NC^{-1}. The maximum torque on the dipole is (a) $0.2 \times 10^{-3} \text{ Nm}$ (b) $1 \times 10^{-3} \text{ Nm}$ (c) $2 \times 10^{-3} \text{ Nm}$ (d) $4 \times 10^{-3} \text{ Nm}$ (iii) Torque on a dipole in uniform electric field is minimum when θ is equal to (a) 0° (b) 90° (c) 180° (d) Both (a) and (c) (iv) When an electric dipole is held at an angle in a uniform electric field, the net force F and torque τ on the dipole are (a) $F = 0, \tau = 0$ (b) $F \neq 0, \tau \neq 0$ (c) $F = 0, \tau \neq 0$ (d) $F \neq 0, \tau = 0$	1+1+1+1
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OR

	An electric dipole of moment p is placed in an electric field of intensity E. The dipole acquires a position such that the axis of the dipole makes an angle with the direction of the field. Assuming that potential energy of the dipole to be zero when $\theta = 90^\circ$, the torque and the potential energy of the dipole will respectively be (a) $pE\sin\theta$, $-pE\cos\theta$ (b) $pE\sin\theta$, $-2pE\cos\theta$ (c) $pE\sin\theta$, $2pE\cos\theta$ (d) $pE\cos\theta$, $-pE\sin\theta$	
30	Read the following paragraph and answer the questions that follow. For most mobile devices, the voltage to recharge the battery is typically 5 volts of direct current. In India, the current supplied to our homes is alternating current at 220V and at a frequency of 50 Hz. Fatima designed a simplified version of a mobile phone charger. She made a circuit using a centre tap transformer and two similar silicon diodes D_1 and D_2 as shown below. Study the diagram below and answer the questions that follow.  1. What is the maximum efficiency of a full wave rectifier? (A) 50% (B) 60% (C) 80% (D) 100% 2. What is the output frequency of a full wave rectifier connected to a 50 Hz AC supply? (A) 50 Hz (B) 100 Hz (C) 150 Hz (D) 200 Hz 3. In a full wave rectifier, what is the function of the transformer? (A) To increase voltage (B) To decrease voltage (C) To provide isolation (D) To filter output 4. What is the main advantage of a full wave rectifier? (A) High voltage regulation (B) High efficiency (C) Low ripple factor (D) All of the above OR In a full wave rectifier, what is the purpose of the load resistor? (A) To increase output voltage (B) To decrease output current (C) To regulate output voltage (D) To provide a path for current flow	1+1+2

SECTION :E

31	Draw a labelled diagram of a moving coil galvanometer and give its principle. Prove that in a radial magnetic field the deflection of the coil is directly proportional to the current flowing in the coil. OR (a) Derive an expression for the force between two parallel current carrying conductors and hence define one ampere. (b) Answer the following: (i) What is the nature of force if current flowing in two wires is in same direction? (i) What is the nature of force if current flowing in two wires is in opposite direction?	2+1+2 3+1+1
32	Draw a curve between mass number and binding energy per nucleon. Give two salient features of the curve. Hence define binding energy. OR What is nuclear force? Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions which you can draw regarding the nature of nuclear force.	2+2+1 1+2+2
33	What are the conditions for sustained interference? What is the effect on the interference pattern observed in a Young's double slit experiment in the following cases: (a) Screen is moved away from the plane of the slits, (b) Separation between the slits is increased, and (c) Widths of the slits are doubled, Give reasons for your answer. OR What is diffraction of light? Discuss single slit experiment for diffraction. Draw the graph to show the relative intensity distribution for a single slit diffraction pattern. Obtain the expression for the width of central maxima.	2+1+1+1 1+2+2

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